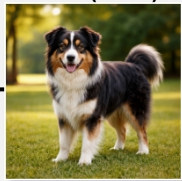


(a) Directed Prior Attraction in CDH

Commonsense (CS) Situation



Question: How many heads does the dog have?  
Candidates: A: 1 B: 2 C: 3 D: 4

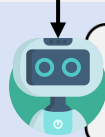
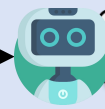


Image supports:  
A: 1



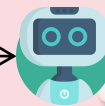
A: 1 ✓

Agreement: the prior is useful

World Prior (text) : A dog normally has one head.

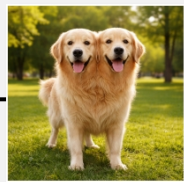


Image supports:  
B: 2



A: 1 ✗

Conflict: the error follows the prior



Question: How many heads does the dog have?  
Candidates: A: 1 B: 2 C: 3 D: 4

Counterfactual (CF) Situation

The same conflict appears in counting, attributes and relations.

(b) CPRC: Adaptive Prior Calibration



Paired CF/CS  
development scores

Offline  
training

Learned  
correction  
calibrator

Frozen for online  
inference

Candidate-aligned correction

Candidate-wise comparison

online  
inference

Prediction  
with image

Prediction  
without image



Test image

Question



No-image  
condition

Single-image online inference

(c) Selective Support Gate

Candidate-aligned proposal

Selective Support Gate

Retain native

Apply correction



A: 1 ✓

Retain  
native answer

Commonsense (CS)  
Situation



B: 2 ✓

Reduce directed  
prior attraction

Counterfactual  
(CF) Situation

Illustrative outcomes

No paired reference image, ground-  
truth label, or CF/CS identity is  
used at test time.